Second-Moment Certificates for Aggregation-Induced GPU Load Blind Spots

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Abstract. Analytical GPU performance models need to distinguish fixed-latency stalls from queueing stalls and account for demand on individual hardware units. ForeSight addresses the latter by recovering per-unit counters with UnitSight. We add a lightweight diagnostic to that per-unit demand vector: for nonnegative loads x_i, $C = \Sigma x_i^2 / \Sigma x_i$ is a certified lower bound on the maximum unit load. In 10,000 seeded synthetic eight-unit log-normal workloads, the mean ratio of actual maximum to certificate was 1.56, compared with 2.90 for the mean-load baseline, with no observed bound violations. A Lean 4 proof checks the two-unit natural-number case and an equal-aggregate counterexample. This CPU-only experiment establishes neither GPU accuracy nor queueing-delay prediction.

1. Motivation and scope

ForeSight observes that queueing stalls vary with pressure on each hardware unit, and UnitSight recovers per-unit counters that ordinary profiler outputs aggregate. The source paper reports an 8.3% kernel-time MAPE across configurations of nine NVIDIA GPUs and 2.0% end-to-end MAPE for its evaluated vLLM workload. We ask a narrower question: given nonnegative per-unit demand, can a scalar concentration statistic provide a cheap, safe diagnostic of load imbalance?

For n units with $x_i \geq 0$ and nonzero $S = \Sigma x_i$, define $C = \Sigma x_i^2/S$, $\mu = S/n$, and $M = \max x_i$. Since $x_i \leq M$, $x_i^2 \leq Mx_i$. Summing yields $C \leq M$. Cauchy–Schwarz gives $S^2 \leq n\Sigma x_i^2$, hence $\mu \leq C$. Thus $\mu \leq C \leq M$. This is a load-domain certificate, not a runtime predictor.

The source arXiv:2611.13762v1 was absent from prior paper artifacts and paper-specific commits before first appearing in the 2026-11-11 daily feed. The `latest` item aliases that snapshot. This audit covers repository history and available session context only; the source v1 text was consulted for this study.

2. Synthetic CPU experiment

The standard-library script uses seed 20261113 and generates 10,000 independent eight-unit vectors with Python random.lognormvariate(0,1). The ratio M/bound measures lower-bound

looseness; closer to one is tighter.

Bound	Mean M/bound	Median	95th percentile
Aggregate mean μ	2.898	2.691	4.786
Second-moment certificate C	1.561	1.569	1.780

Numerical violations: 0/10,000. The vectors (10,0) and (5,5) both have mean 5, but their maxima and certificates are 10 and 5. No GPU kernel, profiler, or queueing simulator was run.

3. Formal scope and limitations

`AggregationGapClaims.lean` compiles with Lean 4.35.0. It proves the equal-sum/different-maximum example and the two-unit natural-number inequality $x^2+y^2 \le \max(x,y)(x+y)$. The general finite-vector argument above is not formalized in Lean. Neither the proof nor experiment validates GPU counters, service rates, or execution times.

The synthetic experiment assumes i.i.d. loads and equal unit capacities. A hardware study should compute the statistic from validated per-unit counters and test held-out predictions. This environment has no available NVIDIA GPU; no hardware result is claimed.

Reproducibility

Run `python3 experiment.py` in this paper directory, then run `~/.elan/bin/lake env lean AggregationGapClaims.lean` from `papers/formal`. The experiment uses only the Python standard library.

References

Wang, Ziwen, Qi Luo, Kunlin Li, Changxi Liu, Dongsheng Wang, and Yun Chen. 2026. “Exploiting What NVIDIA Profilers Hide: Fast and Accurate Analytical GPU Performance Modeling.” arXiv:2611.13762v1. https://arxiv.org/abs/2611.13762v1.